Earth and Planetary Science Letters, 111 (1992) 1–10
Elsevier Science Publishers B.V., Amsterdam

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Paleomagnetic results from the Upper Permian of the eastern Qiangtang Terrane of Tibet and their tectonic implications

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Received July 17, 1991; revision accepted April 6, 1992

ABSTRACT

Paleomagnetic samples were collected from red intercalations within the Upper Permian Tuoba Formation from the eastern Qiangtang Terrane of Tibet. Progressive thermal demagnetization has revealed a prefolding, possibly primary, characteristic remanent magnetization (ChRM), which yields a tilt-corrected mean direction of $D/I = 25.2°/6.7°$ ($\alpha_{95} = 8.6°$). The equatorial paleolatitude for the Qiangtang Terrane indicated by this result and the similarity of Late Permian fauna and flora shared by the Qiangtang Terrane and South China Block (SCB) suggest that the two terranes were in close proximity during the Late Permian. A Late Permian paleogeographic reconstruction involving the major terranes of the present-day East Asia is constructed, in which the Songpan–Garze Sea is surrounded by Laurasia, the North China Block (NCB), the SCB and the Qiangtang Terrane. The Permian development of the Songpan–Garze Sea appears to be related to the rifting of the Qiangtang Terrane off the SCB. Its Triassic evolution may resemble that of the Solomon Sea today in the southwestern Pacific.

1. Introduction

The Tibetan Plateau is composed of several terranes. From north to south, these are the Kunlun Terrane and Songpan–Garze Accretionary Complex, and the Qiangtang, Lhasa and Himalayan terranes, the latter actually being part of the Indian subcontinent (Fig. 1). They are separated from each other by the Jinsha, Bangong–Nujiang and Indus–Zangbo sutures respectively [1–5]. It has been suggested that the Lhasa, Qiangtang, and even the Kunlun terrane were originally rifted from the periphery of Gondwanaland sometime during the late Paleozoic to early Mesozoic [1,5,6–10]. Paleomagnetism can provide quantitative infomation about the paleoposition of these terranes. Paleomagnetic investigations of the Cretaceous and younger rocks from the plateau have yielded good constraints on crustal shortening and rotation for the Lhasa and Qiangtang terranes [9,11–13]. However, studies on the Paleozoic formations have been hampered by the fact that rocks of favorable lithology are scarce and remagnetization becomes a problem that is difficult to resolve. Occasionally, multiphase deformation and metamorphism has obliterated the primary paleomagnetic direction [see 9,14].

In the summer of 1987, we, in collaboration with the geologists of the No.3 Geological Brigade of Yunnan Province who were then mapping the Markam area of eastern Tibet, sampled redbed intercalations in the Late Permian Touba Formation from the eastern Qiangtang Terrane for a paleomagnetic study. This paper reports the results and discusses their tectonic implications.

2. Geology and sampling

Based on stratigraphic and structural features, Chang et al. [5] proposed that the Qiangtang Terrane can be divided, from west to east, into the Qiangtang Block *sensu stricto*, Tanggula–

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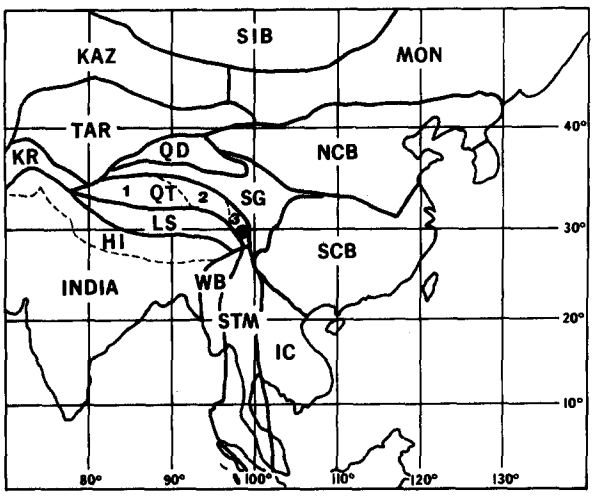


Fig. 1. Schematic tectonic map of China and adjacent area [modified from 48]. Major terranes: *SIB* = Siberia; *KAZ* = Kazakhstan; *MON* = Mongolia; *TAR* = Tarim; *NCB* = North China Block; *SCB* = South China Block; *QD* = Qaidam; *SG* = Songpan–Garze accretionary complex; *QT* = Qiangtang (Qiangtang may be divided into (*1*) the Qiangtang Block *s.s.*,(*2*) the Tanggula–Kaixinling Depression and (*3*) the Qamdo Block); *LS* = Lhasa; *HI* = Himalayan; *KR* = Karakorum; *WB* = West Burma; *STM* = Shan–Thai–Malay; *IC* = Indochina. The large dot denotes the sampling locality of this study. Mercator projection.

Kaixinling Depression and Qamdo Block (Fig. 1). The Qamdo Block is bounded in the east by the Jinsha Suture and in the west by the Taniantaweng orogenic arc. The oldest exposed strata are Lower Ordovician, which are overlain across a pronounced unconformity by a Middle Devonian to Upper Permian littoral and shallow-marine sedimentary sequence with few disconformities in between [5]. The Permian faunas show a strong Tethyan affinity and are similar to those on the SCB. The Upper Permian [1] Tuoba Formation yields a typical Cathaysian flora, equivalent to that of the Longtan Formation of South China [17,18]. The Lower and Middle Triassic are represented by littoral and shallow-marine clastics and carbonates with intercalated intermediate to acidic volcanics. The Upper Triassic is more widespread, consisting, from bottom to top, of non-marine clastics, shallow-marine carbonates, and non-marine clastics with coal seams. Frequent unconformities characterize the sedimentation from end-Permian to Late Triassic times [5]. Conformably overlying the Triassic is a thick redbed sequence of Jurassic–Cretaceous age, which is in turn unconformably capped in places by Eocene to Oligocene red molasse deposits or Pliocene trachytes and clastic rocks.

The Tuoba Formation was described from a type section in the Tuoba area near Qamdo City, about 210 km north of the Haiton section (29.7°N, 98.7°E) where our samples were collected. At the type section, the Tuoba Formation is composed chiefly of blackish/grayish carbonaceous shale, siltstone, and sandstone with coal seams. This succession yields more than 30 plant species, including *Cathaysiodendron acutangulum* (Halle), *Lobatannularia multifolia* Konno and Asama, *Schizoneura manchuriensis* Konno, *Ptychocarpus (Fascipteris) densatus* Gu and Zhi, *Rajahia pseudohemitelioides* Konno, and *Gigantopteris dictyophylloides* Gu and Zhi [18]. This plant fossil association resembles that of the Longtan Formation in South China [17,18]. The underlying Lower Permian formations contain rich fusulinids and corals similar to those occurring in the Qixia and Maokou formations in South China. The overlying Kaxiangda Formation, which is in turn unconformably overlain by andesites and andesitic pyroclastics of the Xiayacun Formation, consists of gray calcareous mudstone and siltstone with marl and sandstone interbeds, and yields the fusulinid *Palaeofusulina*, indicating a latest Permian age (Changxingian) [19]. At the Haiton section 20 km east of downtown Markam, the Permian sequence is closely correlated with that of the Tuoba area. The Lower Permian Lingzhiqiao Formation is composed of more than 1000 m of limestone and bioclastic limestone, which yield fusulinacean index fossils from the *Misellina claudiae* Zone of the Qixian Stage upwards through the *Yabeina–Neomisellina* Zone of the Maokouan Stage, thus

1 The Chinese geologists used to place the Carboniferous/Permian boundary at the top of the Asselian or Wolfcampian and define the Late Permian as limited to the Lopingian subepoch. As a consequence, the conventional Chinese Late Carboniferous includes part of the Early Permian, and the Chinese Early Permian Qixian (Chixian) essentially corresponds to late Early Permian, Early Permian Maokouan corresponds to early Late Permian, and Late Permian Longtanian and Changxiangian corresponds to the late Late Permian that is adopted by scientists from other countries [15,16]. The Chinese usage is maintained throughout this paper unless stated otherwise.

spanning almost the entire Chinese Lower Permian. The Tuoba Formation disconformably overlies the Lingzhiqiao Formation and is in turn disconformably overlain by the Jiangkala Formation, which is made up of andesites and andesitic pyroclastics. The Tuoba Formation is more than 1400 m thick, and primarily consists of gray lithic quartz sandstone, conglomerate and dark gray shale. Few plant fossils have been found from this section, but the occurrence of the fusulinid *Codonofusiella*, the brachiopod *Leptodus* and abundant bivalves such as *Schizodus*, *Euchonodria* and *Aviculopecten* indicates a Late Permian (Longtanian) age. Of particular interest for paleomagnetic study is that there are a few purplish-red mudstone and siltstone interbeds, with a single bed 3–8 m in thickness, in the middle part of the Tuoba Formation. These interbeds outcrop along a series of small sinusoid folds at the sampled section. The fold axes trend NNW–SSE, with limb dips of 20–60°. Ten sites were drilled in the red intercalations in different limbs of the folds. Five to six 2.54 cm diameter cores were taken from each site. A portable rock drill was used for sample extraction and an orientation device mounted with a magnetic compass was used for core orientation.

3. Laboratory techniques

Specimens of standard length were prepared from drilled cores, and thermal demagnetization was used throughout the investigation. Stepwise thermal demagnetization was performed in a TSD-1 Schonstedt thermal demagnetizer, and remanence was measured in an ScT two-axis cryogenic magnetometer at the University of Florida. Samples were subjected to between ten and eighteen heating steps up to apparent temperatures of 685°C. All the equipment was housed in a

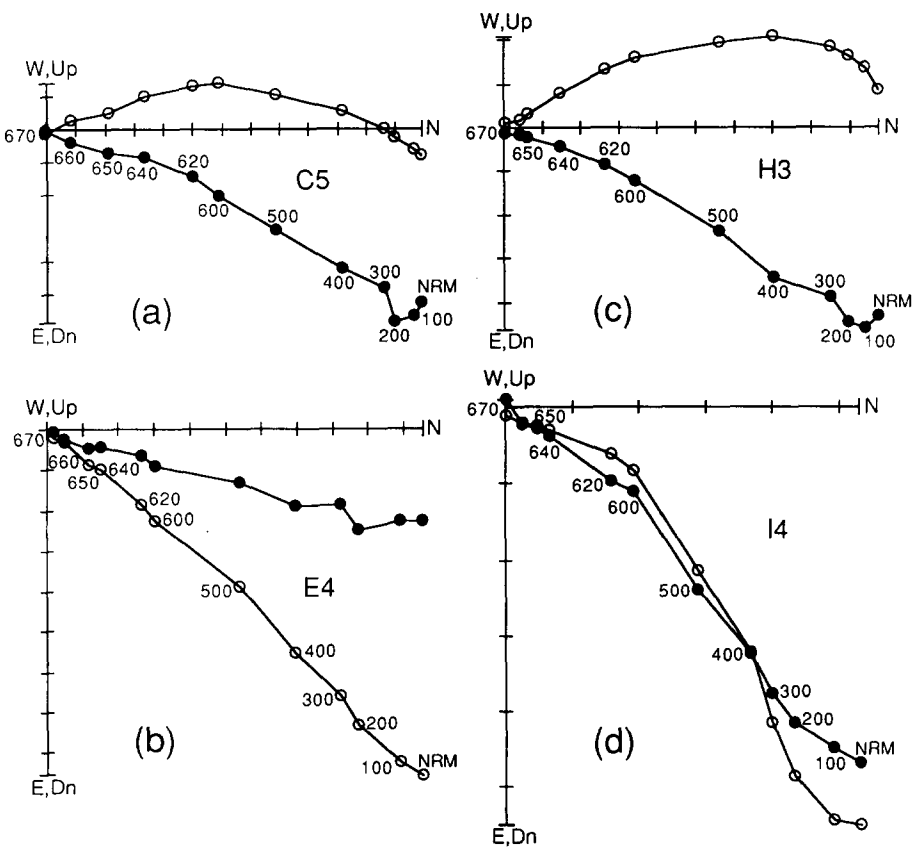


Fig. 2. Representative orthogonal vector plots. Dots are plotted on the horizontal plane, circles on the vertical. Thermal treatment levels are in degrees Celsius. Geographical coordinates.

magnetically shielded room with a residual field of 50–250 nT.

Orthogonal vector plots [20] were constructed to examine the demagnetization behavior of the specimens. Principal component analysis [21] was employed to calculate magnetic directions. Site and formation mean determination used Fisher statistics [22]. Sample directions with maximum angular deviation (MAD) values greater than 15° were rejected for the calculation of site means.

4. Paleomagnetic results

All the specimens measured were strongly magnetized with natural remanent magnetization (NRM) intensity ranging from 7.25 to 18.40 mA/m.

During thermal cleaning, a low temperature component, referred to as the A component, was eliminated below 200°C from most samples. The A component typically accounts for 1–12% of the NRM and exhibited random directions (Fig. 2), probably a viscous remanent magnetization (VRM) acquired during sample storage. With further demagnetization, two more components, which dominate the NRM, were revealed from five sites and from several samples of another two sites (see Table 1): an intermediate temperature component referred to as the B component was generally seen in the temperature range from 300° to 600°C; a high temperature characteristic component, designated as the Ch component, was resolved at above 600°C (Figs. 2a, 2c and 2d). With the exception of a few cases where the B and Ch components are clearly distinguishable (Fig. 2d), these two components often show partially overlapping blocking temperature spectra, which is evident from an arc-shaped transition from the B to the Ch component in the orthogonal vector plots (Figs. 2a and 2c) [23]. In stratigraphic coordinates, both components have similar declinations, but the B component is mostly of steeper inclinations than the Ch component (Table 1). Sites B, E and F and some samples from sites G and J did not show the B component (Fig. 2b), and the characteristic directions were resolved and calculated from the temperature range 200–680°C. Full tilt corrections improve the

TABLE 1

Site statistics (29.7°N/98.7°E)

Site	Bedding	Component	°C	n/N	In Situ D(°)	In Situ I(°)	Tilt D(°)	Tilt I(°)	Corrected α_{95}	Corrected k_2	Pole Lat.°N	Position Long.°E
A	140/20	Ch	610-680	6/6	21.2	-8.6	21.2	9.0	9.7	48.6	57.8	236.1
		B	300-620	6/6	23.7	8.0	20.9	25.7	14.2	23.3		
B	142/20	Ch	400-680	3/5*	19.5	-16.6	21.0	0.5	4.3	834.8	54.4	240.7
C	142/20	Ch	600-680	5/5	13.6	-11.2	14.3	4.6	3.5	489.5	59.4	249.7
		B	300-600	5/5	27.0	11.2	23.6	29.2	5.4	200.9		
D	142/20	Ch	610-685	5/5	19.6	-0.8	18.1	16.0	4.9	243.3	62.6	236.8
		B	200-600	5/5	23.2	18.4	17.9	35.5	9.2	71.0		
E	350/40	Ch	300-680	5/5	21.3	39.2	37.1	13.0	5.1	228.9	48.2	214.8
F	350/40	Ch	200-680	5/5	20.8	42.5	38.9	15.9	5.6	190.5	47.7	211.4
G	150/40	Ch	400-680	5/5	22.3	-17.0	22.7	15.2	8.8	76.2	59.4	230.1
		B	300-600	2/5	32.6	-8.7	29.5	26.6	-	-		
H	150/40	Ch	600-680	5/5	18.0	-21.2	20.7	9.7	6.2	155.8	58.4	236.4
		B	300-600	5/5	31.1	-1.3	24.9	33.1	6.0	166.3		
I	330/50	Ch	610-680	6/6	27.1	19.5	26.2	-23.1	3.3	408.0	41.2	243.6
		B	200-600	6/6	45.1	46.3	49.8	-2.7	5.4	157.0		
J	330/60	Ch	200-670	5/5	9.6	53.3	32.5	4.1	4.9	241.3	48.6	224.5
		B	300-500	1/5	37.1	33.1	39.3	-23.2	-	-		
Formation Mean:												
		Ch		10/10	19.6	7.4	25.2	6.7	8.6		54.4	231.7
					k_1 = 8.3			k_2 = 32.7			A_{95} = 6.1	

Bedding = strike/dip; n/N = number of samples or sites in calculation for means/number of samples or sites measured; D, I = declination and inclination; α_{95}, A_{95} = radius of 95% confidence circle about mean direction and pole, respectively; k_1, k_2 = Fisher precision parameters before and after tilt correction. * Two samples were lost during heating

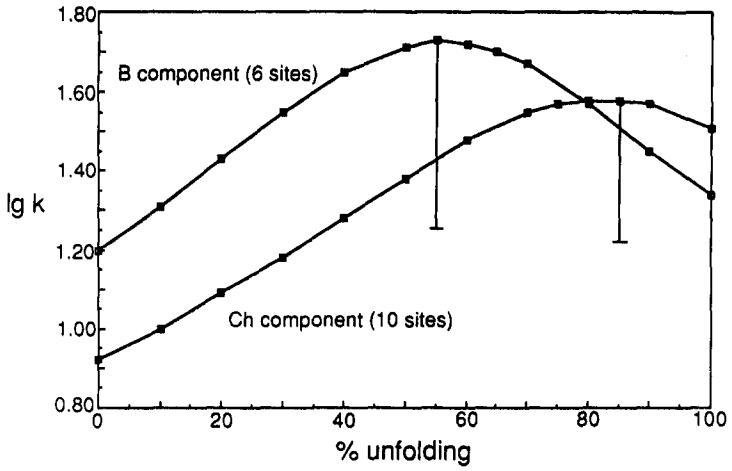


Fig. 3. Variation in Fisher's precision parameter k, with incremental unfolding of the site mean directions of the Ch and B component magnetizations. Vertical bars indicate the 95% confidence limits.

grouping of site mean directions of the B component, which is not statistically significant ($k_2/k_1 = 1.35 < 2.97$, the threshold for 95% confidence when $N = 6$ [24]). However, 55% bedding tilt correction produces a better grouping ($k_2/k_1 = 3.33$) which is significant at the 95% confidence level [24] (Fig. 3). Therefore, the magnetization of the B component was probably acquired during folding. On the other hand, when full tilt corrections are applied, the k_2/k_1 ratio (3.93) for the mean charateristic directions of all the sites is higher than the test value (3.17) for 99% significance [24] (Fig. 3). An 85% unfolding results in a peak in k for the site mean charateristic directions, but this is not significantly different from full tilt correction at the 95% confidence level (Fig. 3). Thus the ChRM passes the fold test and can be regarded as a prefolding magnetization. The tilt-corrected ChRM directions of the ten sites are all north-northeasterly and largely of very shallow positive inclination (Fig. 4). The mean direction for the Tuoba Formation after tilt correction is $D/I = 25.2°/6.7°$ with $\alpha_{95} = 8.6°$, which corresponds to a paleopole at 54.4°N/231.7°E with $A_{95} = 6.1°$.

5. Tectonic implications

As demonstrated above, the ChRM is a prefolding feature. There are, at least, two phases of folding in the study area that formed since the Late Permian. The earlier phase is Late Triassic in age, which is evident from an angular unconformity between the Upper Triassic Jiapeila Formation and older rocks, and is associated with the closure of the Jinsha Suture [5,19]. The later phase is Eocene–Oligocene in age, and is probably related to the collision of India with Asia. It is also reported that the Upper Permian Xiayacun Formation overlies the Kaxiangda Formation above an angular unconformity in the Tuoba area [19], although the contact between the Tuoba and Jiangkala formations appears to be disconformable in the Markam area. Prefolding, possibly Cretaceous, paleomagnetic directions have recently been obtained from the Cretaceous rocks of the Markam area [12]. The paleomagnetic pole computed from the ChRM direction of the Tuoba

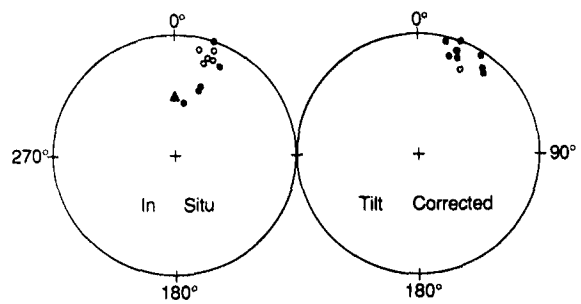


Fig. 4. Equal-area projections of the site mean ChRM directions. Dots represent directions plotted onto the lower hemisphere, circles onto the upper hemisphere. The triangle denotes the present-day magnetic field direction of the Earth at the sampling locality.

Formation is significantly different from those of the Lower and middle Cretaceous formations (Fig. 5), with angular distances being 42.1 ± 19.1° and 32.4 ± 16.7° respectively. Therefore, the ChRM of the Tuoba Formation can be further dated as pre-Cretaceous. Reliable Triassic paleomagnetic data are not yet available from the eastern Qiangtang Terrane, so whether the ChRM of the Tuoba Formation was acquired during the Triassic or Permian cannot be determined. Interestingly, however, the pole position (55.8°N/217.5°E, $A_{95} = 8.8°$) calculated from the synfolding B component at 55% bedding restoration overlaps the pole from the tilt-corrected ChRM direction (Fig. 5), suggesting that the two magnetizations were not greatly separated in time. Moreover, the paleolatitudes indicated by these magnetizations are intimately concordant with the tropical paleoclimatic conditions indicated by the Tuoba flora [17]. Hence it is quite possible that both magnetizations were acquired during latest Permian times.

When paleomagnetic directions are recorded in ancient equatorial latitudes, the choice between normal and reversed polarity is often ambiguous [25]. It is more difficult to tell when only one polarity is sampled. Such is the case for the Permian directions reported here. Nonetheless, we prefer an interpretation of normal polarity for the ChRM, for the following reasons: First, most of the tilt-corrected directions are of downward inclination; second, the Chinese Late Permian (Longtanian plus Changxingian) is actually late Late Permian (mid-late Tatarian) in western usage and is younger than the upper boundary of the Permo-Carboniferous (Kiaman) Reversed Superchron [16,26]; and third, if the reversed polarity is chosen, a rotation of as large as 155° would be required for the sampling site since late Permian times. The resultant paleolatitude for the sampling locality, in line with this polarity assignment, is 3.4°N ± 4.3°.

The paleomagnetic results from the Cretaceous of the Markam area have revealed post-Cretaceous clockwise rotations ranging from 24.5 ± 14.2° to 42.4 ± 17.6° relative to Eurasia [12]. Consistent with this are the preliminary Jurassic results derived from a section 70 km further south, which indicate a rotation of 68.0 ± 22.3° (our own unpublished data). The sampling site

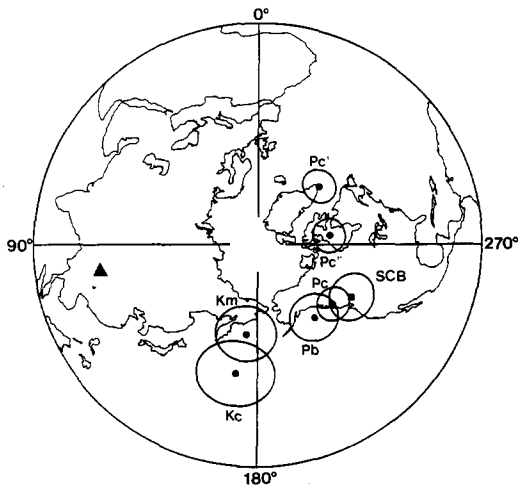


Fig. 5. Paleomagnetic poles and associated 95% confidence circles for the eastern Qiangtang Terrane (dots). *Kc* and *Km* are poles determined from the Lower and middle Cretaceous formations of the Markam area [12]. *Pc* and *Pb* are poles computed from the tilt-corrected ChRM and 55% tilt-corrected B component magnetization of the Tuoba Formation reported in this study. *Pc'* and *Pc''* are the positions of pole *Pc* after correcting for post-Cretaceous rotations indicated by the Lower and middle Cretaceous results [12] respectively. Late Permian pole for the SCB [27] is also shown (square) for comparison. The black triangle denotes the sampling locality of this study. Equal-area projection.

for the Cretaceous is only about 14 km west of that for the Permian, so the same rotations presumably apply to the ChRM of the Permian. Before correcting for the post-Cretaceous rotations, the Late Permian pole for the eastern Qiangtang Terrane is coincidental with the coeval pole for the SCB [27] (Fig. 5). After correcting for the rotations, however, the poles for the two terranes are distinctly different (Fig. 5), suggesting that the eastern Qiangtang Terrane and SCB were not in the present configuration relative to each other during the Late Permian. Note that the paleolatitude for the eastern Qiangtang Terrane is essentially the same before and after correcting for the post-Cretaceous rotations. The difference is only in the paleo-orientation.

A Late Permian (Longtanian) reconstruction of the major terranes of present-day East Asia is presented in Fig. 6. The basic framework of the reconstruction is after Zhao et al. [28], with the following modifications: The Qiangtang Terrane

is positioned in the northern equatorial latitudes and adjacent to the SCB; and the Shan–Thai–Malay and Lhasa terranes are placed midway between the SCB and Australia and between the Qiangtang Terrane and India, respectively. Some authors [e.g. 10] split the Qiangtang Terrane into eastern and western blocks and place them in different paleogeographic positions. It is true that sedimentary rocks of Gondwanan facies have been observed in the Upper Carboniferous and Lower Permian sequence in the western part of the Qiangtang Block *s.s.* [29] whereas the Tanggula–Kaixinling Depression and Qamdo Block yield Carboniferous and Permian faunas similar to those of the SCB [30]. However, the existence of a suspected suture within the Qiangtang Terrane is very uncertain [Yin Jixiang, pers. commun., 1990]. Moreover, it has been shown that the faunal affinity changed from Gondwanan in the Early Permian to Tethyan in the Late Permian in the western part of the Qiangtang Block *s.s.* [29], and that by latest Permian time different parts of the Qiangtang Terrane shared the same Cathaysian flora [17,18,31]. For these reasons we prefer to treat the Qiangtang as an integrated terrane during the Late Permian. The Indochina, Shan–Thai–Malay and Lhasa terranes still lack Permian paleomagnetic data, leaving their accurate paleolatitudinal positions unknown. The Late Permian flora of the Indochina Terrane, however, is similar to that of the SCB [32], indicating that it was geographically close to the SCB during that time. There is strong stratigraphic, sedimentological and paleontological evidence that the Shan–Thai–Malay Terrane was juxtaposed with northwest Australia, probably from the Ordovician to the Early Permian [8]. Middle Permian brachiopods from west and peninsular Thailand, however, show affinities with Indochina and South China [33], implying that the Shan–Thai–Malay

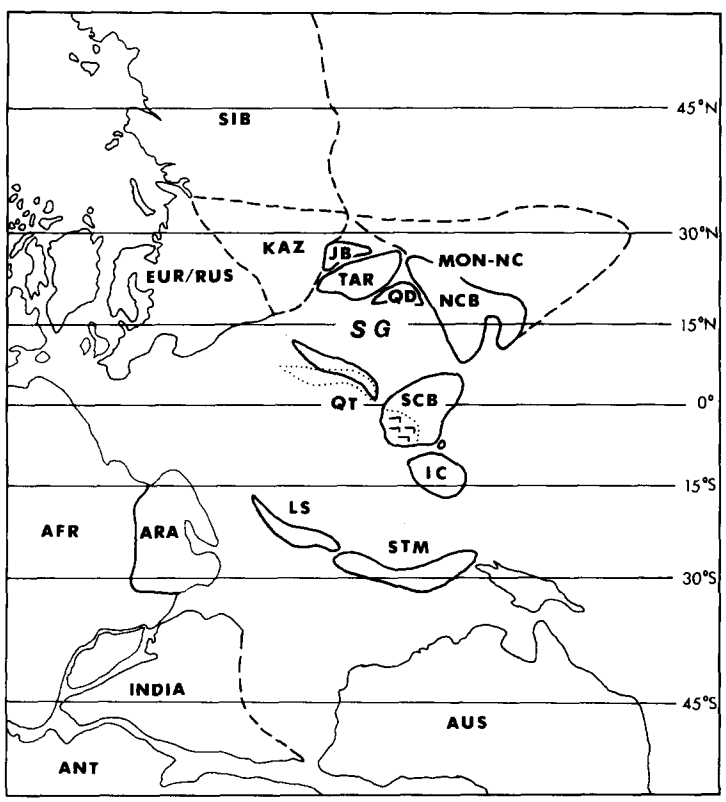


Fig. 6. Schematic Late Permian (Longtanian) geographical reconstruction of the major terranes of present-day East Asia. Terranes: *EUR/RUS* = Europe/Russia; *AFR* = Africa; *ARA* = Arabia; *ANT* = Antarctica; *AUS* = Australia; *JB* = Junggar. See Fig. 1 for remaining abbreviations. Solid and dashed outlines of the Qiangtang Terrane correspond to the rotation-corrected poles *Pc'* and *Pc''* respectively in Fig. 5. The ornamented southwestern part of the SCB shows the Emeishan basaltic province.

Terrane had probably rifted away from Australia. Similarly, a parsimony analysis of endemism of marine faunas of the Tibetan plateau shows a switch of the Lhasa Terrane from strong ties with the Himalayan Terrane during the earliest Permian to closer links with the Qiangtang Terrane during the late Early Permian (Moukouan) [34], suggesting that by the Late Permian the Lhasa Terrane had left Gondwanaland. This is also a view advocated by Metcalfe [8].

It now seems clear that the Songpan–Garze Sea had formed by the Late Permian, surrounded by the Tarim Terrane (and probably Qaidam Terrane as well) to the north, the NCB to the northeast, the SCB to the southeast, and the Qiangtang Terrane to the southwest (Fig. 6). Its development was probably related to rifting of the Qiangtang Terrane off the SCB. This is evident from the Permian andesites and basalts that occur in the Qiangtang and Kunlun terranes and that show intraplate geochemical characteristics reflecting eruptions in rifts on attenuated continental lithosphere [35]. Likewise, studies on major and trace element geochemistry of the flood-type Emeishan basalt in the western SCB have revealed intraplate compositions with no component related to subduction, and a compositional variation trend from continental in the east of the basaltic province to oceanic in the west [36]. Permian basaltic eruption also developed to the west of the SCB in the Songpan–Garze region, where volcanics often coexist with marine sediments and pillow structures are common [36–38]. The broad synchroneity and the intraplate nature of these basalts indicate that they were products of the same rifting event. To the west of the Longmenshan fracture zone, which marks the northwestern boundary of the SCB [39], pillow basalts of Late Permian age immediately overlie Early Permian platform carbonates and clastics correlating with those on the SCB, and are in turn overlain by the Triassic flysch [40]. In addition, a continental basement beneath the thick Triassic flysch pile in the Songpan region (Songpan Massif) is inferred from aeromagnetic survey data [38,39]. This suggests that part of the floor of the Songpan–Garze Sea might consist of pieces of the SCB that have been broken off.

Subduction of the Songpan–Garze Sea essentially commenced during the Triassic along both the northern and southwestern margins [7,30], and it culminated in the Middle to mid-Late Triassic. The northwestern margin of the SCB appears to have been a passive continental margin [7,41] and to have been related to the consuming margins of the Songpan–Garze Sea probably through transform faults [7]. A large quantity of clastic material was shed primarily off the Laurasian highlands to the northwest and dumped into this sea [42], thus creating probably the largest flysch basin in the world. The sediments in the northeastern portion of the sea were monotonous turbiditic clastics (the Xikang Group), whereas they were interbedded with abundant volcanics in the southwestern part (the Dege Group). Such an asymmetry resulted from the fact that there was more than one subduction zone, and therefore more than one volcanic arc in the southwest [7]. The double-sided subduction of the Songpan–Garze Sea under the Tarim–(Qaidam)–North China terranes in the north and the Qiangtang Terrane in the southwest, in conjunction with clockwise rotation of the SCB and Qiangtang Terrane [28,30], effectively consumed the floor of the Songpan–Garze Sea. By the latest Triassic, the northern and southern terranes and associated accretionary complexes initially collided [7,30]. However, collision must have continued into the Jurassic, as is evident from widespread granodioritic intrusions in the Songpan–Garze system [7] and granitoid intrusions in the northern Qinling Mountains [15]. Paleomagnetic results recently obtained from the SCB and NCB also favor a final Jurassic closure of the Qinling Suture [43,44]. Sengor and Hsu [45] [see also 46] regard the Songpan–Garze system as a typical example of a "non-appositional" suture because the internal deformation and thickening of its large accretionary complexes stopped convergence before the colliding northern and southern continents could be apposed. If there were pieces of continental basement in the Songpan–Garze Sea, however, they would certainly play a role in bringing about such "non-apposition" of the sutures. The Solomon Sea in the southwestern Pacific [47] is a possible modern analog of the Triassic Songpan–Garze Sea. For a detailed account of the Triassic evolution of the Songpan–Garze accretionary complex, the reader is referred to [7].

6. Conclusion

A prefolding ChRM isolated from the Upper Permian Tuoba Formation indicates equatorial paleolatitudes for the eastern Qiangtang Terrane. The similarity in marine faunas and, especially, the (Cathaysian) flora shared by the Qiangtang Terrane and SCB further suggests the close geographic proximity of these terranes during the Late Permian. Such a paleogeographic proximity, as well as geological and geochemical data of the Permian volcanic rocks from the Qiangtang Terrane, SCB and Songpan–Garze system, appears to support the idea that the Permian development of the Songpan–Garze Sea was probably due to the rifting of the Qiangtang Terrane off the SCB. The Triassic evolution of the Songpan–Garze Sea probably resembles that of the Solomon Sea today in that double-sided subduction coupled with rotation of bordering continents is effectively suturing the region from west to east [47].

Acknowledgements

This is part of a cooperative project between the Institute of Geology of the Academia Sinica and the Department of Geology of the University of Florida. This study was supported by the National Natural Science Foundation of China (K.N.H.) and by the Division of Earth Sciences, National Science Foundation, grant EAR 8816968 (N.D.O.). We thank Professors Yin Jixiang and Chang Chengfa for helpful discussion about the geology of the Tibetan plateau. Critical reviews by David Rowley and Vincent Courtillot improved the paper.

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